

Can navigate the terminal: ls, cd, whoami, pwd.

ls = list files

cd = go into folder

pwd = current directory

whoami = current user

I can do it.

Can create, delete files / folders: mkdir, rm, cp, mv, touch, nano.

mkdir = make a new directory

rm = remove files or directories

cp = copy files or directories

mv = moves files

touch = creates new empty file

nano = opens a file in a text editor

I can do it

Can use the following basic terminal commands: cat, wc, uniq, sort.

cat = display contents of a file

wc = counts lines and characters in a file

uniq = removes duplicated lines

sort = sort lines in a file alphabetically or numerically

I can do it

Can create a new repository in your preferred Git provider.

Step 1: Click new repository

Step 2: Choose settings and name

Step 3: click create

I can do it.

Can perform basic Git operations: clone, add, commit, push, pull.

git clone = clones a repository to a local machine

git add = moves a file to the staging area and marking it for the next commit

git commit = Saves changes with a message

git push = uploads local changes to the remote repository

Git pull = fetches and merges changes from a remote repository

I can do it:

Can list the main hardware components of a computer, their functions and the Von Neumann architecture.

CPU

RAM

STORAGE

Von Neumann architecture:

Program + data stored in the same memory

CPU fetches then decodes and executes instructions

Can explain how computers work, starting from hardware all the way to software.

Hardware in a computer executes machine code binary.

The Operating System (OS) manages the hardware and allows the programs to run.

The software are coded so they can comply into the machine code

I can do it

Can talk about processes in operating systems.

It has its own memory space while it can run at the same time with others and its managed by the OS.

Can talk about different number representations such as:

- Binary

Its a base 2 and its used internally by computers

- Hexadecimal

Its a base 16 and its a compact representation of binary

- Decimal

Its a base 10 and its used for to represent data so humans can read it

I can do it

Can bring up real-world use cases for different number representations.

Binary = machine level operations

Hexadecimal = addresses and colors

Decimals = everyday calculations

I can do it

Can explain different charsets like ASCII and Unicode and how they differ.

ASCII = foundational character encoding 128 characters. It allows computers to store text in 7 bit integers.

I can do it

Knows about different types of databases and their use cases.

SQL = relational database. Stores data and organizes in tables of rows and columns.

NoSQL = non-relational database. Stores data in systems unstructured and where the data changes frequently.

Can setup a new MySQL database and connect to it.

```
CREATE DATABASE mydb;
```

I can do it

Can create DDL statements to create tables.

```
CREATE TABLE users (  
id INT PRIMARY KEY,  
name VARCHAR(100)  
);
```

I can do it

Can create DML (SELECT**, **INSERT**) statements.**

```
INSERT INTO users (id, name) VALUES (1, 'Mario');  
  
SELECT * FROM users;
```

I can do it

Can use **WHERE clause to filter results.**

```
SELECT * FROM users WHERE name = 'Mario';
```

Can define Primary Keys.

Cannot be null

Unique identifier for each row

Understands how Foreign Keys work.

Reference another table's primary key

Used to link tables

In **SELECT** statements, can use (with moderate help):

- **LIMIT, ORDER BY, GROUP BY**

```
SELECT name FROM users
```

```
ORDER BY name
```

```
LIMIT 5;
```

- aggregate functions like **COUNT**, **SUM**, **AVG**, **MIN**, and **MAX**

```
SELECT COUNT(*), AVG(id) FROM users;
```

- pattern matching with **LIKE** and wildcards

```
SELECT * FROM users WHERE name LIKE 'D%';
```

- **JOIN** to combine data from multiple tables

```
SELECT users.name, orders.id
```

```
FROM users
```

```
JOIN orders ON users.id = orders.user_id;
```

Can create inner, left and right joins after looking up the syntax.

Inner join:

```
SELECT * FROM a INNER JOIN b ON a.id = b.id;
```

Left join:

```
SELECT * FROM a LEFT JOIN b ON a.id = b.id;
```

Right join:

```
SELECT * FROM a RIGHT JOIN b ON a.id = b.id;
```

Can create DDL statements to create tables with constraints: PRIMARY KEY, AUTO_INCREMENT, FOREIGN KEY, UNIQUE, NOT NULL.

```
CREATE TABLE users (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    email VARCHAR(100) UNIQUE NOT NULL  
);
```

I can do it

Can create DML (UPDATE, DELETE) statements.

```
UPDATE users SET name = 'Mario' WHERE id = 1;
```

```
DELETE FROM users WHERE id = 1;
```

Can create a pull request.

Step 1: push branch

Step 2: click compare& pull request

I can do it

Understands different Git workflows such as GitHub Flow.

Create branch

Work → commit

Open pull request

Review → merge

Can solve a merge conflict.

git add .

git commit

I can do it

Can write YAML.

name: John Doe

age: 25

student: true

Understands what GitHub Actions are and can breakdown workflows into runners, jobs, and steps.

Runner is machine executing workflow

A job is a group of steps

A step is an individual task

Can give use cases for GitHub Actions.

CI/CD pipelines

Testing code automatically

Deployment

I can do it

Understands different cloud service models: IaaS, PaaS, SaaS.

IaaS = infrastructure (VMs)

PaaS = platform (deploy apps)

SaaS = ready-to-use software

Can deploy a web application to Azure App Service.

Microsoft Azure:

Create App Service

Connect repo

Deploy

Understands the flow from pushing, GitHub Actions running, building the project and deployment to Azure.

1. Code push
2. GitHub Actions runs
3. It builds
4. Deploys to Azure

Adding a database to a Spring Project.

1. Add dependency
2. Configure datasource
3. Connect via JDBC

Setting up a database in Azure.

1. Create Azure SQL / MySQL
2. Configure access
3. Connect via connection string

Can set up a database in Azure and connect it to an Azure App Service project with a guide.

I can do it

Is able to go through scenarios that can cause concurrency problems in databases.

Issues that occur when multiple transactions access the same data at the same time.

Can explain what ACID is and why it solves concurrency problems for databases.

ACID is a set of properties (Atomicity, Consistency, Isolation, Durability) that ensure reliable and safe database transactions.

Is aware of the possibility to define transactions in SQL, JDBC and atomic structures.

Transactions in SQL and JDBC is a way to group multiple database operations so they either all succeed or all fail together.

Atomic structures are operations or components that are executed completely or not at all without interruption and its important for thread safety.